

Assignment 7 Statistics

Each complete question and its options are followed by the correct answer.

Question 1: What is the primary purpose of statistics? A. To store data B. To collect, analyze, interpret, and communicate data C. To increase the size of a dataset D. To convert all data into numbers

Answer: B. To collect, analyze, interpret, and communicate data

Question 2: Which of the following is an example of quantitative data? A. Eye color B. Blood group C. Height D. Country

Answer: C. Height

Question 3: Which of the following is an example of categorical data? A. Age B. Weight C. Temperature D. Blood group

Answer: D. Blood group

Question 4: A population refers to: A. A small subset of observations B. The entire group being studied C. Only the observations used for training D. Only numerical observations

Answer: B. The entire group being studied

Question 5: A sample is: A. The entire population B. A subset of the population C. Always larger than the population D. A single variable

Answer: B. A subset of the population

Question 6: Why is sampling commonly used? A. Studying the entire population may be expensive or impractical B. Samples always provide perfect results C. Populations cannot be measured D. Sampling eliminates all errors

Answer: A. Studying the entire population may be expensive or impractical

Question 7: Which measure represents the arithmetic average of a dataset? A. Median B. Mode C. Mean D. Range

Answer: C. Mean

Question 8: What is the mean of: 10, 20, 30, 40, 50 A. 20 B. 25 C. 30 D. 35

Answer: C. 30

Question 9: The median is: A. The largest value B. The smallest value C. The middle value after arranging the data D. The most frequent value

Answer: C. The middle value after arranging the data

Question 10: What is the median of: 3, 7, 8, 10, 15 A. 7 B. 8 C. 10 D. 15

Answer: B. 8

Question 11: What is the median of: 2, 4, 6, 8 A. 4 B. 5 C. 6 D. 7

Answer: B. 5

Question 12: The mode represents: A. The average value B. The middle value C. The most frequently occurring value D. The largest value

Answer: C. The most frequently occurring value

Question 13: Which dataset has a clear mode? A. 1, 2, 3, 4, 5 B. 2, 2, 3, 4, 5 C. 1, 2, 3, 4, 5, 6 D. 10, 20, 30, 40

Answer: B. 2, 2, 3, 4, 5

Question 14: Which measure is generally more sensitive to extreme values? A. Mean B. Median C. Mode D. Frequency

Answer: A. Mean

Question 15: Consider: 10, 12, 13, 14, 100 Which measure would be most affected by the value 100? A. Mean B. Median C. Mode D. Minimum

Answer: A. Mean

Question 16: If a very large outlier is added to a dataset, the mean will generally: A. Decrease B. Increase C. Remain exactly the same D. Become zero

Answer: B. Increase

Question 17: For a highly skewed dataset containing extreme values, which measure may better represent the typical observation? A. Mean B. Median C. Variance D. Range

Answer: B. Median

Question 18: A dataset has mean = 50 and median = 50. What can we conclude? A. The distribution must be perfectly normal B. The distribution is necessarily symmetric C. Mean and median happen to be equal D. There are no outliers

Answer: C. Mean and median happen to be equal

Question 19: What does dispersion describe? A. The center of the data B. How spread out the data are C. The number of observations D. The data type

Answer: B. How spread out the data are

Question 20: Which of the following is a measure of dispersion? A. Mean B. Median C. Standard deviation D. Mode

Answer: C. Standard deviation

Question 21: The range of a dataset is: A. Mean – Median B. Maximum – Minimum C. Maximum + Minimum D. Median – Minimum

Answer: B. Maximum – Minimum

Question 22: What is the range of: 5, 8, 10, 15, 20 A. 10 B. 15 C. 20 D. 25

Answer: B. 15

Question 23: If all values in a dataset are identical, the standard deviation is: A. 0 B. 1 C. Equal to the mean D. Infinite

Answer: A. 0

Question 24: A larger standard deviation generally indicates: A. Less spread B. Greater spread C. Smaller mean D. More observations

Answer: B. Greater spread

Question 25: Two datasets have the same mean. Dataset A has standard deviation 2, while Dataset B has standard deviation 20. Which statement is most appropriate? A. Dataset A has greater variability B. Dataset B has greater variability C. Both have identical variability D. Standard deviation cannot compare datasets

Answer: B. Dataset B has greater variability

Question 26: Variance is based on: A. Absolute values from zero B. Squared deviations from the mean C. The maximum value only D. The median

Answer: B. Squared deviations from the mean

Question 27: Why are deviations squared when calculating variance? A. To make all deviations positive and emphasize larger deviations B. To make the mean zero C. To remove the observations D. To reduce the dataset size

Answer: A. To make all deviations positive and emphasize larger deviations

Question 28: If the variance of a dataset is 25, its standard deviation is: A. 5 B. 10 C. 25 D. 625

Answer: A. 5

Question 29: If the standard deviation is 4, the variance is: A. 2 B. 4 C. 8 D. 16

Answer: D. 16

Question 30: Which statement is correct? A. Variance and standard deviation always have the same units B. Standard deviation is the square root of variance C. Variance is the square root of standard deviation D. Variance is always smaller than standard deviation

Answer: B. Standard deviation is the square root of variance

Question 31: A distribution describes: A. How values are arranged or occur in a dataset B. Only the largest value C. Only the average D. The number of columns

Answer: A. How values are arranged or occur in a dataset

Question 32: A symmetric distribution has: A. No mean B. Similar shape on both sides of its center C. Only positive values D. No variation

Answer: B. Similar shape on both sides of its center

Question 33: In a perfectly symmetric distribution, the mean and median are generally: A. Very different B. Approximately equal C. Always zero D. Undefined

Answer: B. Approximately equal

Question 34: A distribution with a long tail toward larger values is described as: A. Left -skewed B. Right -skewed C. Uniform D. Symmetric

Answer: B. Right -skewed

Question 35: A distribution with a long tail toward smaller values is: A. Right -skewed B. Left -skewed C. Uniform D. Normal

Answer: B. Left -skewed

Question 36: In a positively skewed distribution, which relationship is generally expected? A. Mean > Median B. Mean < Median C. Mean = 0 D. Median = 0

Answer: A. Mean > Median

Question 37: A normal distribution is generally: A. Symmetric and bell -shaped B. Always skewed C. Uniform D. Discrete only

Answer: A. Symmetric and bell -shaped

Question 38: In an ideal normal distribution, the mean, median, and mode are: A. Different B. Approximately equal C. Always zero D. Undefined

Answer: B. Approximately equal

Question 39: The standard deviation of a distribution determines: A. Its center only B. Its spread C. Its number of observations D. Its maximum value

Answer: B. Its spread

Question 40: Increasing the standard deviation of a normal distribution generally makes the distribution: A. Narrower B. Wider C. Shift to the right D. Shift to the left

Answer: B. Wider

Question 41: Decreasing the standard deviation while keeping the mean unchanged makes a normal distribution: A. Wider B. Narrower C. More skewed D. Shifted right

Answer: B. Narrower

Question 42: A z -score indicates: A. The original value of an observation B. How far an observation is from the mean in standard deviation units C. The number of observations D. The variance of the dataset

Answer: B. How far an observation is from the mean in standard deviation units

Question 43: If an observation has a z -score of 0, it is: A. One standard deviation above the mean B. At the mean C. One standard deviation below the mean D. At the maximum

Answer: A. One standard deviation above the mean

Question 44: A z -score of +2 means that the observation is: A. 2 units above the mean B. 2 standard deviations above the mean C. 2 standard deviations below the mean D. 2 units below the mean

Answer: B. 2 standard deviations above the mean

Question 45: A z -score of -3 means that the observation is: A. 3 units below the mean B. 3 standard deviations below the mean C. 3 standard deviations above the mean D. Equal to the mean

Answer: B. 3 standard deviations below the mean

Question 46: Correlation measures: A. The average of a variable B. The strength and direction of association between two variables C. The number of observations D. The variance of one variable only

Answer: B. The strength and direction of association between two variables

Question 47: A correlation coefficient close to +1 indicates: A. Strong positive linear association B. Strong negative linear association C. No linear association D. Causation

Answer: A. Strong positive linear association

Question 48: A correlation coefficient close to -1 indicates: A. Strong positive association B. Strong negative linear association C. No association D. Identical variables

Answer: B. Strong negative linear association

Question 49: A correlation coefficient close to 0 generally indicates: A. Strong positive linear relationship B. Strong negative linear relationship C. Little or no linear relationship D. Perfect causation

Answer: C. Little or no linear relationship

Question 50: If height and weight have a positive correlation, it means: A. Increasing height necessarily causes weight to increase B. Taller individuals tend to have higher weight C. Every tall person is heavier D. Height and weight are identical variables

Answer: B. Taller individuals tend to have higher weight

Question 51: Which statement about correlation is correct? A. Correlation always implies causation B. Correlation does not necessarily imply causation C. Correlation can only be positive D. Correlation can only be used for categorical variables

Answer: B. Correlation does not necessarily imply causation

Question 52: The median divides an ordered dataset into: A. Two equal parts B. Three equal parts C. Four equal parts D. Ten equal parts

Answer: A. Two equal parts

Question 53: The first quartile (Q1) represents approximately: A. 10th percentile B. 25th percentile C. 50th percentile D. 75th percentile

Answer: B. 25th percentile

Question 54: The third quartile (Q3) represents approximately: A. 25th percentile B. 50th percentile C. 75th percentile D. 90th percentile

Answer: C. 75th percentile

Question 55: The interquartile range (IQR) is: A. $Q1 + Q3$ B. $Q3 - Q1$ C. $Q3 \times Q1$ D. $Q1 / Q3$

Answer: A. $Q1 + Q3$

Question 56: Why is IQR useful? A. It measures the spread of the middle 50% of the data B. It measures the total number of observations C. It always equals the standard deviation D. It identifies the mean

Answer: B. It measures the total number of observations

Question 57: Random sampling means: A. Selecting only the largest observations B. Giving members of the population a chance of being selected C. Selecting observations manually D. Selecting only convenient observations

Answer: B. Giving members of the population a chance of being selected

Question 58: A sample that does not properly represent the population may lead to: A. Sampling bias B. Zero variance C. Perfect prediction D. Normal distribution

Answer: A. Sampling bias

Question 59: Increasing the sample size generally: A. Makes estimates less reliable B. Can make estimates more stable C. Eliminates all sources of bias D. Guarantees a normal population

Answer: B. Can make estimates more stable

Question 60: Which of the following is most likely to produce a biased sample? A. Randomly selecting people from the entire population B. Selecting participants only from one convenient location C. Using a properly designed random sampling method D. Repeating random sampling several times

Answer: B. Selecting participants only from one convenient location

Question 61: A dataset has: Mean = 50 Median = 50 Standard deviation = 2 Which statement is most reasonable? A. The values are generally close to 50 B. The values are extremely spread out C. The mean must be an outlier D. There is no variation

Answer: A. The values are generally close to 50

Question 62: A dataset has: Mean = 50 Standard deviation = 30 What does this suggest? A. The data have relatively high variability around the mean B. All values are close to 50 C. All observations equal 50 D. The dataset has no variance

Answer: A. The data have relatively high variability around the mean

Question 63: Two datasets have identical means and different standard deviations. What does this tell us? A. They have exactly the same distribution B. Their centers are the same, but their variability differs C. Their medians must be different D. Their sample sizes must be different

Answer: B. Their centers are the same, but their variability differs

Question 64: A dataset has a strong positive correlation between two variables. Which conclusion is safest? A. One variable definitely causes the other B. The variables have a strong positive linear association C. The variables must have identical values D. There is no relationship

Answer: B. The variables have a strong positive linear association

Question 65: Which statistic is generally most appropriate for describing the center of highly skewed data with extreme outliers? A. Mean B. Median C. Variance D. Standard deviation

Answer: B. Median